A certified re-check that every cubic bipartite graph on at most 68 vertices has a cycle of length 4, 8 or 16

(not new; a witness-checked recomputation)

Xavier Callens (SocrateAI Lab)

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AI-assistance disclosure and audit status. This preprint, its C and Python code, its claim ledger and the audit were produced by AI agents (Claude, Anthropic) under the author's direction; the author is responsible for the content. **This version was independently audited by a different AI agent instance (`AUDIT.md`, `ACCEPT_WITH_CORRECTIONS`), not by a human.** The audit's nine corrections are applied in this text (Section 8). No human audit exists; it is owed (Section 10). There is no Lean in this paper.

Abstract

We re-check, with exhaustive search and machine-checkable certificates, that **every simple cubic bipartite graph on at most** 68 **vertices contains a cycle of length** 4, 8 **or** 16**.** Equivalently, no connected linear v_3-configuration with $v \leq 34$ points avoids Berge 4- and 8-cycles. We follow the growth search of Tranquilli (arXiv:2608.02675), whose certified range stops at 58 vertices, in two root modes: *triangle-rooted* (his mode; for orders 62–68 it needs the literature premise $n(3, 10) = 70$, read by us only through the Exoo–Jajcay cage survey) and *arbitrary-root* (no triangle assumed, no cage premise). Two oracles for the forbidden-cycle test give identical decision-transcript hashes at every cap. Route 2 caps 29–34 (orders 58–68) and route 1 caps 30–35 close with 0 completions. Certificate streams are accepted by the producer's two checkers and by an independent auditor's own C checker; the cap-34 arbitrary-root stream was regenerated by the auditor and has the same sha256 (`0112b808...fad5`). Controls reproduce Tranquilli's Table 2 at cap 29, the canonical class counts $1, 1, 3, 10, 31, 229, 2036$ for $v = 7..13$ (equal to the nauty `genbg` sets), and positive and negative filter-off controls. **This is not a new bound.** Ripatti (BAIKAL/MOTOR 2017 abstract) claims, without a certificate, that the smallest cubic graphs without cycles of length $4, 8, 16$ have 78 vertices, which covers every cubic graph on at most 76 vertices, and Ducoffe–Dumitru (arXiv:2609.28594) cover bipartite orders up to 66 by SAT without a proof log. We know of no earlier witness-checked computation for these orders; the contribution is a recomputation with certificates and an independent audit. Tier [C] (exhaustive search plus witness checkers, no kernel check); the coverage lemma is argued in prose and not formalised.

What this is NOT. It is **not** a new bound (see Section 6), **not** progress on the Erdős–Gyárfás conjecture, **not** a statement about non-bipartite, non-cubic or larger graphs, and **not** formalised in Lean or any proof assistant. Independent here means independent *code*; the method (Tranquilli's search) is shared by producer and auditor.

Markers. [L]: a cited source, taken as stated. [C]: executed code with an argument written down but not machine-checked. [X]: a recorded absence, supporting nothing by itself. There is no Tier A row. Identifiers `EGC-·-nnnn` refer to `ledger/ledger.json`.

1 Statement and equivalent forms

Statement. Every simple cubic bipartite graph on at most 68 vertices contains a cycle of length 4, 8 or 16. A bipartite cubic graph has two sides of equal size v, so order ≤ 68 means $v \leq 34$. A graph with no 4-cycle is the Levi (point–block incidence) graph of a linear v_3-configuration (Tranquilli Prop. 3, Lemma 5): v points and v blocks of size 3, any two points on at most one block. A Levi 8-cycle is a Berge 4-cycle and a Levi 16-cycle is a Berge 8-cycle (Lemma 6). Hence the statement is equivalent to: *no connected linear v_3-configuration with $v \leq 34$ has neither a Berge 4-cycle nor a Berge 8-cycle* (a disconnected counterexample contains a connected one of smaller order, so connectedness costs nothing).

Context. Erdős and Gyárfás conjectured that every graph of minimum degree at least 3 has a cycle whose length is a power of two. For cubic bipartite graphs of order at most 68 our statement is stronger than what the conjecture needs (it uses only the lengths $4, 8, 16$, not 32 or 64), but it is a finite computation about small orders; **nothing is claimed about the conjecture itself.**

2 Method

Search (Tier [C]). `data/code/cb_search.c.txt` is a new C implementation written only from the text of Tranquilli §3.1, Lemmas 5–7 and Prop. 10 (`EGC-L-0001`); the producer did not open his repository. The search installs root blocks, processes the least unfinished point p and proposes a new block $\{p, q, r\}$ in lexicographic order after the last block through p with least point p; q, r range over eligible old labels ($> p$, degree < 3, not yet collinear with p) or the next one or two fresh labels, within the cap. A proposal is rejected, in this order, if it is *structural* (q and r already share a block), if it creates a *C8* (a simple Levi path of length 6 between two old points of the block), or a *C16* (length 14). A completion is recorded when every introduced point has degree 3. There is no isomorphism pruning.

Two root modes.

- **Route 1, triangle-rooted** (Tranquilli's mode): roots $\{0,1,3\}, \{1,2,4\}, \{0,2,5\}$ plus $\{0,4,6\}$ (orbit 1) or $\{0,6,7\}$ (orbit 2). This is complete for configurations with a Berge triangle (Levi 6-cycle). A C_4- and C_8-free cubic bipartite graph has girth 6, 10 or ≥ 12; girth ≥ 10 needs at least 62 vertices (Moore, his Lemma 8), girth exactly 10 needs at least 70 (the premise $n(3, 10) = 70$, Tier [L], `EGC-L-0002`: O'Keefe–Wong 1980, computer assisted, read by us *only* through the Exoo–Jajcay Dynamic Cage Survey, EJC DS16 v3, p. 7 Table 1 and p. 10 §2.1.6; the original was read by nobody), and girth ≥ 12 needs at least 126. So for order ≤ 60 route 1 needs only the Moore bound, and for orders 62–68 it additionally needs $n(3, 10) = 70$.
- **Route 2, arbitrary root**: the single root block $\{0, 1, 2\}$, no triangle assumed. The same coverage argument applies because $\{0, 1, 2\}$ is the lexicographically first block through 0 under first-occurrence labelling. **This route needs no Moore or cage premise.**

Two oracles. Oracle A computes depth-first endpoint sets, cached per state. Oracle B joins complete lists of half-paths (length 3 or 7) per pair, uncached. Both must give the same counters *and* the same FNV-1a hash of the ordered decision transcript; they do at every cap and root mode where both were run (Tables 1, 2).

Witness checkers. A certificate stream records, for every proposal, either acceptance (then the checker recurses) or a rejection with its witness: for C8/C16 a simple path of length 6 or 14 in the pre-insertion Levi graph between two distinct introduced members of the block. Three checkers read these streams: the producer's `cb_check.py` (Python) and `cb_check_fast.c` (C), and the auditor's independent `aud_check.c`. They search for nothing; they re-derive the proposal schedule as a sorted, filtered candidate list (a different formulation from the nested loops of `cb_search`), check every structural rejection by collinearity, check every witness path, recurse into expansions, refuse any completion, and recompute all counters against the END line. Accepting a proposal needs no proof (it only enlarges the tree), so soundness rests on the rejections being witnessed and the schedule being the Prop. 10 schedule.

Acceptance condition (audit correction 6). The producer's two checkers read the cap, the orbit and the filter flags from the stream's own header. An honest stream with C6 pruning (`-c6`) or an honest cap-20 stream is accepted by them. An accepted stream therefore supports the claim *only if* the checker's reported cap and orbit equal the claim, its flags are empty and its C6 count is 0. All recorded check outputs satisfy this (route 2 cap 34 has $c_8 > 0$ and $c_{16} > 0$, so neither filter was off, and $c_6 = 0$). The auditor's `aud_check` hard-codes cap and orbit and rejects both such streams.

Coverage lemma (Tier [C], not formalised). Tranquilli's Prop. 10, re-derived by the producer and, separately, by the auditor (AUDIT.md §1; `EGC-C-0001`). In outline: keep the invariant that labels are consecutive, every point below p is finished, and the inserted blocks with least point p form a lexicographic initial segment. Every uninserted block through p is $\{p, q, r\}$ with $q, r > p$; at most two remain; blocks with two old points or one old and one new point sort by their least old point, blocks with two new points sort last; linearity forbids two remaining blocks sharing a point; new labels are handed out in that order; so the next target block is lexicographically after the last inserted one and is among the schedule's candidates; it passes the structural, C8 and C16 tests because a sub-configuration of a linear configuration without Berge 4- or 8-cycles has none either; connectedness forces every point to be introduced. **An argument is not a proof in a checker.** It is validated empirically (Section 3) up to $v = 13$ (arbitrary root) and $v = 14$ (triangle roots) against nauty, and beyond that it rests on the text.

3 Controls

Run before any headline number (`PREREG_ITEM.md`). Everything below is the producer's run unless marked (A) for the auditor's repetition.

id	control	result
P1	Cap-29 counters vs Tranquilli Table 2 (states, attempted, structural, C8, C16, completions), orbit 1 and orbit 2	orbit 1: 1,405/106,964/7,184/63,526/34,850/0; orbit 2: 20,088/1,655,404/113,012/1,208,473/313,832/0. Identical (also (A))
P1	19-block states (paper §3.3: 337)	$46 + 291 = 337$
P1	Oracle A vs B at cap 29; Python checker on the cap-29 certificates	same counters and hashes; accepted
P2	C8 and C16 filters off: canonical classes (nauty `labelg`, bipartition kept) for $v = 7..13$	$1, 1, 3, 10, 31, 229, 2036$ in both root modes (A: also)
P2b	Set equality with `nauty-genbg -c -d3:3 -D3:3 -Z1`, $v = 7..13$	all 7 sets equal. (A): arbitrary root $v \leq 13$ and triangle roots $v = 14$ (21,398 classes) equal `genbg`
P2c	`genbg` plus an independent DFS cycle filter, no connectivity condition in (A)	0 graphs without C_4, C_8, C_{16} for $v \leq 16$ (producer, connected) and $v \leq 15$ (A: $1, 1, 3, 10, 31, 229, 2036, 21399, 245342$ graphs, every one with a C_8). $v = 16$ was not repeated by (A)
P3	C16 filter off, cap 19, triangle roots	completions exist: 16 labelled, 1 class, C_4/C_8-free with a C_{16}
P3b	Same, arbitrary root	128 labelled leaves, 1 class; see below
N1	C8 filter off, C16 on, cap 13	completions appear: the Fano plane ($v = 7$)
N2	C8 filter on, cap 8; Heawood graph	0 completions; with the filter off Fano ($v = 7$) and 8_3 ($v = 8$) appear and the checker accepts. Heawood: girth 6, 21 eight-cycles, C8 test rejects its last block
N3	Tamper tests: truncated stream, flipped counter, corrupted C8 witness, wrong-length C16 witness, C16 relabelled structural, deleted record, trailing data, inserted fake completion	all 8 rejected by `cb_check.py` and `cb_check_fast`. (A): 11 further kinds rejected by all three checkers

P3b and the number 128 **(audit correction 7).** Tranquilli writes of 128 symmetric 19_3-configurations. The single class found with C16 off at cap 19 is self-dual: its colour-preserving automorphism group has order $57 = 19 \cdot 3$ (order 114 as an uncoloured Levi graph), the 114 rooted choices fall into two orbits, and the auditor's independent search finds the same single class in all three root modes (128 labelled leaves with root $\{0, 1, 2\}$, 16 with orbit 2, 0 with orbit 1). This supports, but does not prove, that the paper's 128 are labelled copies of one configuration. The equality $128 = 128$ is therefore not evidence that no class was missed.

Not matched. Tranquilli's Table 4 leaf counts $(1, 4, 44, 496, 7840, \dots)$ differ from both implementations: our triangle-rooted leaves are $1, 4, 44, 511, 5190, 60231, 752362$, our arbitrary-root leaves $2, 8, 100, 1360, 17968, 263214, 4027664$, and the auditor's search gives exactly the producer's counts in every mode. The canonical class sets agree with nauty, so this is a counting convention of that paper's generator, not a coverage gap, but it remains *unexplained*. Only the canonical counts were preregistered.

Deviation in the controls. The first N2 run used cap 7, which `cb_search` refuses (minimum cap 8); the `N2_cap7_*` files record that failure and N2 was rerun at cap 8. `heawood_c8.py` was fixed twice for networkx-version issues before it ran (re-hashes logged in `freeze_sha256.txt`).

4 Results

Columns: states, attempted proposals, structural, C8, C16 rejections, completions. The last column of each row shows whether, in order, oracle B (producer), the auditor's independent search and the producer's binary rebuilt by the auditor reproduce the counters *and* the transcript hash (`Y` yes, `-` not run). The tables are generated by `ledger/gen_tables.py` from the recorded JSON files, which also asserts equality.

4.1 Route 1: triangle-rooted, caps 30–35

cap (order)	states	attempted	structural	C8	C16	compl.	transcript hash	B/aud/reb
30 (60), orbit 1	2 247	197 766	13 106	115 032	67 382	**0**	`56ad49915df0282d`	`YYY`
30 (60), orbit 2	32 153	2 838 956	187 816	1 992 302	626 686	**0**	`ba1f0883aefd766d`	`YYY`
31 (62), orbit 1	4 011	403 911	26 046	233 736	140 119	**0**	`b1e590c5ea776d84`	`YYY`
31 (62), orbit 2	52 541	5 024 144	321 347	3 394 496	1 255 761	**0**	`5aed270f87a161f9`	`YYY`
32 (64), orbit 1	7 112	742 895	46 650	411 221	277 913	**0**	`4d6b30d329a06b26`	`YYY`
32 (64), orbit 2	82 021	8 648 052	528 841	5 649 649	2 387 542	**0**	`a347451c04d122c4`	`YYY`
33 (66), orbit 1	12 377	1 417 878	86 955	737 586	580 961	**0**	`ed7f846980005e34`	`YYY`
33 (66), orbit 2	126 017	14 700 719	858 762	9 284 478	4 431 463	**0**	`3a03e5ad73585470`	`YYY`
34 (68), orbit 1	23 143	2 972 750	180 011	1 424 785	1 344 812	**0**	`653a538e27efd809`	`YYY`
34 (68), orbit 2	202 602	25 227 209	1 432 521	15 207 779	8 384 308	**0**	`3d4a09ae6dca2b8b`	`YYY`
35 (70), orbit 1	51 974	7 282 166	441 386	3 199 099	3 589 708	**0**	`f36c872dddf7fe0f`	`YY-`
35 (70), orbit 2	353 357	47 941 559	2 696 312	27 114 667	17 777 224	**0**	`42744fc11246012d`	`YY-`

Table 1: Route 1. Every row: oracle B hash equal to oracle A, certificate accepted by the producer's Python checker and by `aud_check` (the 12 certificates are listed in the appendix). Cap 35 (order 70) covers only the girth-6 part and is *not* a statement about order 70: the three $(3, 10)$-cages lie outside the triangle-rooted search.

The growth factors in total states were 1.60 (cap 30), 1.64 (31), 1.58 (32), 1.55 (33), 1.63 (34) and 1.80 (35); the preregistered kill criterion (projection above 48 core-hours for cap 34) never fired. Cap 34 took 5.4 CPU-s (orbit 1) and 43.3 CPU-s (orbit 2) with oracle A including certificate writing, and 2.0 s and 13.5 s with oracle B.

4.2 Route 2: arbitrary root, caps 29–34

cap (order)	states	attempted	structural	C8	C16	compl.	transcript hash	B/aud/reb
29 (58), arbitrary root	656 769	62 551 453	4 349 082	41 493 681	16 051 922	**0**	`7ff1613205286506`	`YYY`
30 (60), arbitrary root	1 221 044	123 345 322	8 394 650	82 188 835	31 540 794	**0**	`ec5db4e8c1ac7531`	`YYY`
31 (62), arbitrary root	2 286 787	250 478 360	16 640 536	166 568 141	64 982 897	**0**	`6cc50daf1ccf1608`	`YYY`
32 (64), arbitrary root	4 178 622	500 012 963	32 156 507	328 949 906	134 727 929	**0**	`38267459e04ce7ab`	`YYY`
33 (66), arbitrary root	7 508 537	974 464 041	60 683 233	624 694 002	281 578 270	**0**	`1b4fb3765c2ee344`	`YY-`
34 (68), arbitrary root	13 798 102	1 899 509 410	115 596 123	1 154 879 407	615 235 779	**0**	`76e8805d44433718`	`YY-`

Table 2: Route 2 (no triangle assumed, no premise). Single-core CPU-s without a certificate, oracle A / oracle B (producer): cap 29: 5.3 / 29.5; 30: 11.2 / 59.4; 31: 20.5 / 121.9; 32: 39.1 / 246.6; 33: 75.3 / 505.6; 34: 149.8 / 1062.8. For caps 33–34 the auditor's rebuilt-binary comparison was not run in this table form; the cap-34 certificate run of the producer's binary (hash `76e8805d44433718`) is in `data/audit/out/r34_certrun.json`.

The cap-34 arbitrary-root certificate (the one that carries the headline). The stream is tens of GB and was never stored. The producer regenerated it with oracle A and streamed it through a FIFO into `cb_check_fast`: accepted, all counters equal (`data/main/R_cap34_checkfast.json`); stream sha256 `0112b8085b969bcfa75335fdb2aedc5d4b5b0e80c6dddf2d5d9ead8f5a18fad5` (`data/logs/main/R_cap34_stream_sha256.txt`). The certificate run's transcript hash (`76e8805d44433718`) equals that of the plain oracle-A run and of oracle B. The *auditor* then regenerated the stream from a binary rebuilt from the shipped source (sha256 equal to the freeze value) and streamed it into `aud_check` (cap 34,

orbit 0 and no flags hard-coded): **ACCEPT**, all counters equal, hash `76e8805d44433718`, stream sha256 identical to the producer's. Wall time 1 h 53 min (19:00–20:53 UTC), producer CPU 3,605 s.

A Python check that never finished. A Python check of an identically generated stream (`cb_check.py`, core 3) started at 2026-10-31 14:41 UTC and was stopped unfinished at 17:34 UTC after the literature finding (Section 6); it had read at least 17.5 GB without a rejection (it aborts at the first error), and its partial-stream hash was deleted, so there is nothing to compare. **No Python check of the route-2 cap-34 stream ever completed, by either party;** the stream has been checked by two C checkers (the producer's `cb_check_fast` and the auditor's `aud_check`).

Certificates are outside the bundle. The 12 route-1 streams are 134,625,781 bytes of gzip in all (128.4 MiB); they are kept at `/home/callensxavier_gmail_com/EG-CB68_scratch/certs/` on the producer's machine and are *not* in the repository or the Zenodo bundle. Their names, sizes and sha256 (gzip and decompressed) are in the appendix and in `data/certs_inventory.json`; the sha256 values were recomputed for this paper (by streaming decompression) and agree with the producer's `data/logs/main/cert_sha256.txt` and with the auditor's own hashes. **They can be regenerated:** the search is deterministic, `cb_search CAP ORBIT A -cert FILE` writes the same stream (same transcript hash), and the auditor regenerated the one stream that was never stored and obtained the recorded sha256. The two cap-29 control certificates are in the same scratch folder (`P1_cap29_o1/2.cert.gz`, hashes in `cert_sha256.txt`).

5 What the auditor did

A different AI agent instance (Claude Opus) audited the lane on 2026-10-31 (`AUDIT.md`, verdict **ACCEPT_WITH_CORRECTIONS**; not a person). It used its own code in `/tmp` outside the repository (sources shipped in `data/audit/`), cores 2 and 3 only (at times more than two processes shared them, which inflated wall times).

- **Independent search** `aud_search.c`: reproduces every counter and every transcript hash of both routes (route 1 caps 30–35, route 2 caps 29–34). Producer code was not reused. The schedule is the Prop. 10 schedule written as a sorted candidate list: *independent code, not an independent method*; a canonical-augmentation or SAT method would be independent.
- **Rebuilt producer binaries**: `cb_search.c` and `cb_check_fast.c` compile to the frozen sources' sha256 values (`8928c27b...`, `868e8ffc...`; the sources shipped here hash to them) and reproduce route 1 caps 30–34 and route 2 caps 29–32 and the route-2 cap-34 certificate run. We also compiled the shipped `.c.txt` sources and reproduced cap 29, orbit 1, for this paper.
- **Independent checker** `aud_check.c`: accepts the 12 route-1 streams and the regenerated route-2 cap-34 stream; rejects 11 tamper kinds (and the producer's `cb_check.py`, `cb_check_fast` reject them too).
- **Full enumeration independent of any search**: `nauty-genbg -d3:3 -D3:3 -Z1` for $v = 7..15$ (orders up to 30; no connectivity condition), piped into `aud_g6cyc`: none is free of C_4, C_8, C_{16}; filter sanity checks: Heawood, Tutte–Coxeter and Petersen give a C_8, the cube a C_4, C_{10} none (negative control), C_{16} a C_{16}. The counts equal the known v_3-configuration counts (Betten–Brinkmann–Pisanski) including disconnected ones.
- **Coverage validation** against nauty, filters off: arbitrary root, cap 13 and triangle roots, cap 14 (Section 3).
- **Reading** of both producer checkers against the soundness conditions of Section 2; a re-derivation of the coverage lemma for both root modes; recomputation of the growth factors.

Producer $\neq$ verifier. Satisfied by this audit, in the following sense and no further: an independent search reproduced all counters and hashes and an independent checker accepted the certificates. It is an AI audit, not a human one.

6 Priority: this is not new

The preregistered novelty claim (order 68 new) is **killed**. A same-day re-search on 2026-10-31, after the runs, found:

- **Ripatti**, BAIKAL/MOTOR 2017 one-page abstract (`isem.irk.ru/conferences/mopt2017/Abstracts/BAIKAL_2017_p`) "the smallest cubic graphs without 4,8,16 cycles have 78 vertices". This covers all cubic graphs on at most 76 vertices, bipartite or not, hence our statement and the planned extension to order 72. The method is canonical augmentation with "probabilistic checking of every node"; no certificate is reported, and no full paper was found. Exoo's 78-vertex graph is not bipartite.
- **Ducoffe–Dumitru** (arXiv:2609.28594 v1): bipartite graphs of minimum degree 3 and order ≤ 66 by SAT modulo symmetries; they forbid $4, 8, 16, 32$ (and 64 at orders 64–66), assume minimality reductions that are vacuous for cubic graphs, and "did not ask the solver for a resolution proof". Within cubic bipartite graphs our 4/8/16 statement is formally stronger than theirs at orders 60–66, but not stronger than Ripatti's claim.
- **Garcia** (arXiv:2609.04686) quotes $54 \leq f(4) \leq 78$ (Exoo) and, like Tranquilli and Ducoffe–Dumitru, does not cite Ripatti.

Wording (audit correction 1). *We know of no earlier witness-checked computation for cubic bipartite orders* 60–68*. Ripatti (2017, abstract, uncertified) and Ducoffe–Dumitru (2026, SAT without a proof log) cover these orders. This recomputation was independently audited (AI audit, not human).* This is a statement of absence, recorded as Tier [X] (`EGC-X-0001`): the searches were an alphaXiv search and a web search on 2026-10-28 (which missed Ripatti) and on 2026-10-31, and one query each for OEIS and House of Graphs by the auditor. It is not a priority claim. Tranquilli's certified result stops at 58 vertices.

Consequence for the work done. The exploratory extension to arbitrary-root caps ≥ 35 (preregistered only by Addendum 1) was stopped at 17:34 UTC on 2026-10-31 without any result, because orders 70 and 72 would also have been re-checks; its outputs are not used (`data/logs/stopped_runs.txt`). The file `data/logs/main/R_cap35_stream_sha256.txt` belongs to that stopped run and is not a hash of any used stream.

7 The chain, with tiers

claim	tier	ledger
Coverage lemma (both root modes)	[C]	`EGC-C-0001` (not formalised)
Route 1: no completion, caps 30–35, certificates accepted	[C]	`EGC-C-0004`
Route 2: no completion, caps 29–34, cap-34 stream accepted by two C checkers	[C]	`EGC-C-0005`
Graph statement, order ≤ 68, by route 2 (no premise)	[C]	`EGC-C-0007` (headline)
Graph statement, order ≤ 60, by route 1 plus Moore	[C]	`EGC-C-0008`
Graph statement, orders 62–68, by route 1	[C] conditional on [L]	`EGC-C-0008`, `EGC-L-0002`
$n(3, 10) = 70$	[L]	via a survey only
Not new: Ripatti, Ducoffe–Dumitru	[L]	`EGC-L-0003`

No order is new as a bound. Nothing is claimed beyond 68 vertices, outside cubic bipartite graphs, or about the full conjecture.

8 Audit corrections applied

The audit (`AUDIT.md`, §6) required nine corrections. They are applied as follows. The producer's `data/RESULT.md` and `data/result.json` are kept *unmodified* (their hashes are recorded); `data/result_audited.json` is the audited-fields copy of `result.json`, and this paper supersedes `RESULT.md`.

1. **Novelty wording**: replaced by the sentence in Section 6 ("we know of no earlier witness-checked computation ...").
2. **Producer ≠ verifier**: the pending statements in `RESULT.md` §5, §6.9, `result.json` `producer_ne_verifier` and `tiers` are superseded: satisfied by the audit (independent search, independent checker, same cap-34 stream sha256). In `result_audited.json`.
3. **Python check of the route-2 stream**: kept as "stopped unfinished" (Section 4), with the addition that the auditor's independent C checker `aud_check` accepted an identically generated stream (same sha256 `0112b808...fad5`).
4. `result.json`, `logs/make_result.out`: the recorded hash `e3b0c442...b855` is the sha256 of the *empty* file (it was hashed while `make_result.py` was still writing it, and a file cannot hash itself). The entry is removed in `result_audited.json`.
5. **Git-ignored logs**: nine `logs/*.out` files are git-ignored (`.gitignore:77 *.out`), including `rcert34_fast.out`, which is cited as evidence for the route-2 check, and so are *local only* in the lane directory. They are shipped in this paper's `data/logs/` as `*.out.txt` copies (several are empty, as recorded).
6. **Checker acceptance condition**: stated in Section 2.
7. **P3b and self-duality**: Section 3 replaces "our interpretation of a numerical coincidence" by the self-duality statement (colour-preserving group of order 57).
8. **Addendum 1 timestamp**: `PREREG_ADDENDUM_1.md` says "written ... ∼18:10 UTC", but its recorded hash timestamp is 17:53:31 UTC (`data/logs/prereg_item_sha256.txt`). The written-at time in its header is wrong; the hash is earlier than any extension run either way. Reported as a deviation (Section 10).
9. **Paper number**: the method source is Tranquilli, arXiv:2608.02675; **2609.04686 is Garcia**, not Tranquilli. The audit brief had it wrong; the producer's `RESULT.md` and this paper cite both correctly, and the Zenodo metadata uses both numbers with the relation "references".

9 Claim ledger

`ledger/build_ledger.py` files 13 claims: 3 at Tier [L] (`EGC-L-0001..0003`), 8 at [C] (`EGC-C-0001..0008`, the headline being `-0007`) and 2 at [X] (`EGC-X-0001`, the absence of an earlier certified computation; `EGC-X-0002`, what was not run). There is no Tier A row. Each carries a hashed evidence blob in `ledger/evidence/`. The Elenchus gate (`tools/ledger.py` of the SocrateAI Elenchus checkout) ran with `ledger/run_gate.py`; the verbatim output is in `ledger/gate_output.txt`: "13 claims checked, no findings", exit code 0. The gate checks bookkeeping (tier ceilings, dependency closure); it licenses nothing about the mathematics.

10 Limitations and deviations

- **Not new.** A re-check of uncertified computations (Section 6); the classification rests on a one-page abstract of Ripatti 2017 and on short searches.
- **AI audit only; no human audit.** No person has read the code or the argument.
- **Unverified, from the audit.** (a) The coverage lemma is argued in prose, not formalised; checked against nauty for every connected configuration with $v \leq 13$ (arbitrary root) and $v \leq 14$ (triangle roots), and beyond that it rests on the argument. (b) The auditor's search shares the Prop. 10 schedule and the kind of endpoint-set path test with the producer: independent code, not an independent method. (c) $n(3, 10) = 70$ was not read in the original; only route 1 at orders 62–68 depends on it, route 2 does not. (d) There is no full text of Ripatti 2017. (e) `genbg` at $v = 16$ was not repeated by the auditor (the producer's $v \leq 16$ run is same-author). (f) The Python check of the route-2 cap-34 stream never completed; two C checkers stand in. (g) Tranquilli's Table 4 leaf counts remain unexplained.
- **Tier [C] is the ceiling.** No Lean, no kernel check, no statement lock.

- **Scope.** Cubic bipartite graphs of order at most 68 only. The route-1 cap-35 run covers the girth-6 part of order 70 and says nothing about the $(3, 10)$-cages. A statement about orders ≥ 70 is not made.
- **Deviations from the preregistration.** (1) The parent assigns the search code to a Sonnet-tier agent; the producer was Opus. (2) Arbitrary-root mode (route 2) was added before the freeze, and the C checker `cb_check_fast` after it (logged in `freeze_sha256.txt`, tamper-tested). (3) $n(3, 10) = 70$ read only in a survey. (4) N2 cap $7 \to 8$ (Section 3). (5) `PREREG_ITEM.md` was written by the producer and hashed before the first run; Addendum 1 (the extension) was written after the timing probe, hashed 2026-10-28 17:53:31 UTC (header says ∼18:10, see correction 8); Addendum 2 moved the extension's wall-clock deadline to 2026-11-01 14:45 UTC after an external API spend-limit outage (about 2026-10-28 18:30 to 2026-10-31 14:40 UTC), written before any extension run (hashed 14:41:15 UTC). (6) An exploratory girth-≥ 10 run (C6 filter, C16 off, arbitrary root) was started and stopped at cap 33 for cost, before its result; only its cap-31 output (0 completions) exists, in a session log, and it is not used. (7) The repository guards `antigravity_guard.py` and `test_rigor_guard.py` both exit 1 on pre-existing files elsewhere; neither flags an EG-CB68 file (`data/logs/guard_*.txt`); `pytest tests/` was not run. (8) The novelty check failed (Section 6). (9) Compute: about 7 core-hours of the 48 core-hour box, an estimate from process wall times, cores 2 and 3 only.
- Not a physical claim of any kind.

11 Reproducibility

Repository AutoevolveAI, branch `worktree-openai-math-discovery`; HEAD when this paper was written: `e0fbdba6f0bae6249785ed25b1224a620e77f91a`; the lane records were committed by the producer in `abe4718ed552e3c77b7207cadb8095d998e02602`, the audit, this paper and the auditor's material were uncommitted working-tree files at that time; nothing was pushed or uploaded. Copies of all inputs are in `data/`: the producer's `RESULT.md`, `result.json`, preregistration files and `AUDIT.md`; `code/` (sources; C sources are stored as `.c.txt` and shell scripts as `.sh.txt`; compiled binaries are not shipped); `main/` and `controls/`; `logs/` (the git-ignored `*.out` logs as `.out.txt`); and `audit/` (the auditor's `aud_search.c`, `aud_check.c`, `aud_g6cyc.c`, their run scripts and small outputs; the tampered streams, 3.4 MB each, and the auditor's copy of the Ripatti PDF are not shipped). No third-party full text is included.

```
gcc -O3 -march=native -x c -o cb_search       data/code/cb_search.c.txt        # sha256 of the source: 8928c27b...
gcc -O3 -march=native -x c -o cb_check_fast   data/code/cb_check_fast.c.txt    # sha256 of the source: 868e8ffc...
./cb_search 29 1 A                         # cap 29, orbit 1, oracle A: 1405 / 106964 / 7184 / 63526 / 34850 / 0 (Tranquilli
     Table 2)
mkfifo f ; ./cb_search 34 0 A -cert f > R34.json & ./cb_check_fast < f     # route 2, cap 34, streamed through a FIFO (about
     1-2 h, tens of GB)
./cb_search 34 2 A -cert T_cap34_o2.cert ; python3 -I data/code/cb_check.py T_cap34_o2.cert   # route 1, orbit 2, cap 34
python3 -I ledger/make_cert_inventory.py ; python3 -I ledger/gen_tables.py     # inventory and tables (the first needs the
     external certs)
python3 -I ledger/build_ledger.py ; python3 -I ledger/run_gate.py
```

A The external route-1 certificates

Kept outside the bundle at `/home/callensxavier_gmail_com/EG-CB68_scratch/certs/` (regenerable, see Section 4). For each file the gzip sha256 (line "gz") and the sha256 of the decompressed stream (line "plain"), with sizes. All hashes: producer's `cert_sha256.txt`, the auditor's, and the recomputation for this paper agree; the auditor's `aud_check` and the producer's `cb_check_fast` accept every stream with cap and orbit as named, flags empty and $c_6 = 0$ (`data/audit/out/certs/`).

file (gzip)	bytes (gz)	bytes (plain)
`T_cap30_o1.cert.gz`		
gz 5984fbb99dd51e2cabc0ddd373d62b0c407c50736e3e22492f6292b949a1e5bf	250 781	6 383 362
plain 7bef71ce306f464ca84f74b09b962d3668750431f3f9e5b60ef852f6d9e67b3d		
`T_cap30_o2.cert.gz`		
gz 80f1e1b8d49f659c33dcff63352df9a8769fd93ff8b8f9ba21cdb5b833fdd9d6	2 732 383	83 992 668
plain d61874d9d298753d046fd4a4bf778561e2eb10a87e55f23f63d2574cd289588a		
`T_cap31_o1.cert.gz`		
gz 22b7dfae1e3de4b9a242851dea7ef659448795a05998026fe08c6e066192c6c1	492 954	13 176 730
plain f9ff763d22217ba9d9ff57a45a52f95d6511863f50457712b0e1a7a1573b7b79		

file (gzip)	bytes (gz)	bytes (plain)
`T_cap31_o2.cert.gz`		
gz 29879a217e96a7774fd41e89be9e3b69df3f0b7f4b9c19464e30d7468d7b7d5c	4 968 652	152 849 671
plain 10724687aece5f672c182c867427c460ce47d5d4d05be21ebd309e1468f320f5		
`T_cap32_o1.cert.gz`		
gz 8f4969983d71a8750ba00198ba2ccbf0d1f4457b1ec072758c44fb86b6242170	893 476	24 775 350
plain 8ea1ce43a026bf8cae9a702608d5249eeecfc37d76218f4e46660614e8011140		
`T_cap32_o2.cert.gz`		
gz e37b6c7b0539e4c64f6c6c7ea2d67573b1ecefcc0cbd4ee6bb66c87531469ab1	9 142 297	269 675 736
plain 386b44245c9b78185a9e1583a0f6b9b231959d76f56b83b52c6e16a813b215ae		
`T_cap33_o1.cert.gz`		
gz 2d710a1456741489ddc203992bde95db830a5163e7123a95c3700b05505df165	1 726 375	48 613 168
plain d16943d41ac0b99dfa126cf23d8ecaa3afbd2f9c17c965a748125535c0bfb820		
`T_cap33_o2.cert.gz`		
gz b43308713dcb65024be9608145cc300c4803659163796602a9dbf530a746ad36	15 906 246	469 014 095
plain c604eeb2876a8357727ed64b8f14b48629cc89874eff077a5cb10d1739dbdac0		
`T_cap34_o1.cert.gz`		
gz b5b2f64fdc1c09e02cc765dfd2c31d7a9fe329589fad8c1f2d08529ae045cdbe	3 807 306	105 107 594
plain f3e7c3199cde3e25fc4fba78c349e86929fe6e0ea8dad144007de2f33ca9a4e9		
`T_cap34_o2.cert.gz`		
gz 3f189eb350f03dd42a2d767157c6f0b3d98ea4509828caa5a33009c46c88308c	28 312 241	825 299 502
plain 5195c0efc9cc3cbec675849bd7b6315263407358f1dffd1916725710e73d59c0		
`T_cap35_o1.cert.gz`		
gz 0a3e7727f3487bd46d1d4ba041c0c441acbea86bd90fa6e26d948b10f2b8b07a	9 552 228	264 821 167
plain cfd97e591993d31b8037e101b7005a4248c548b70f24e5eda7183a2a1d039a26		
`T_cap35_o2.cert.gz`		
gz 743ae6b5d50c90a6d2deba0ca837f3bed440eb2e2c53e6e68d2445ae5ed5e0d4	56 840 842	1 615 100 437
plain d07b0012946387042c6300afd86318b323d30a634d70b0188fb3cc15434eeede		

References

[1] J. Tranquilli, arXiv:2608.02675 (v1), certified search for cubic bipartite counterexamples to the Erdős–Gyárfás conjecture (title as quoted in the audit: "A 60-Vertex Lower Bound for Cubic Bipartite Counterexamples…"). The method source: Prop. 3, Lemmas 5–9, Prop. 10, Tables 2 and 4. ([L])

[2] D. Garcia, "Small graphs without power-of-two cycles…", arXiv:2609.04686. Read for $f(4)$ context; cites "Exoo gave $54 \le f(4) \le 78$". Not the method source. ([L])

[3] Ducoffe, Dumitru, arXiv:2609.28594 (v1, 2026). Bipartite graphs of order ≤ 66 by SAT, no proof log. ([L])

[4] A. V. Ripatti, "Faster approach for exhaustive search over graphs with certain properties", abstract, BAIKAL/MOTOR 2017, `isem.irk.ru/conferences/mopt2017/Abstracts/BAIKAL_2017_paper_138.pdf`. One page; claims 78 as the smallest order of a cubic graph without 4-, 8-, 16-cycles. ([L])

[5] G. Exoo, R. Jajcay, Dynamic cage survey, Electron. J. Combin. DS16 (v3, 2013); used for $n(3, 10) = 70$ (p. 7 Table 1, p. 10 §2.1.6). ([L])

[6] M. O'Keefe, P. K. Wong, The smallest graph of girth 10 and valency 3, J. Combin. Theory Ser. B 29 (1980) 91–105; *not read.* ([L])

[7] B. D. McKay, A. Piperno, Practical graph isomorphism II, J. Symbolic Comput. 60 (2014) 94–112; `nauty genbg`, `labelg`, `countg` as packaged for the machine; used for the controls and the audit's enumeration.

[8] A. Betten, G. Brinkmann, T. Pisanski, Counting symmetric configurations v_3, Discrete Appl. Math. 99 (2000) 331–338; the counts quoted by the auditor; bibliographic details from memory, not checked, and the paper was not re-read here.